

The Meta-Cognitive Autopilot: A Unified Control-Theoretic Framework for Biomimetic Evasion of Advanced Behavioral Classifiers via Chaotic Systems, Stochastic State-Space Estimation, and Meta-Control Barrier Functions

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Abstract

Modern Web Application Firewalls (WAFs) increasingly deploy machine learning-based behavioral classifiers that analyze temporal patterns and pointer movement kinematics to distinguish automated agents (web bots) from human users. This paper introduces a unified, biomimetic, closed-loop evasion framework, termed the *Meta-Cognitive Autopilot*, designed to bypass advanced behavioral classifiers with optimal efficiency and safety. The framework is developed in three progressive layers: (1) an outer-loop biological generator that couples a one-dimensional Logistic Map chaotic temporal scheduler with a minimum-jerk Cubic Bezier trajectory generator; (2) an inner-loop cognitive state estimator that models WAF alertness as a stochastic differential equation (SDE) and dynamically tracks it using a covariance-bounded Kalman filter; and (3) an outer-loop meta-controller governed by a proportional-integral-derivative (PID) law that dynamically optimizes the filter's uncertainty lower bound to eliminate estimation lag. Safety guarantees are mathematically enforced via Control Barrier Functions (CBFs) computed through Quadratic Programming (QP). We evaluate our framework in a dynamic threat wind-tunnel over 10,000 continuous sequential requests. Empirical results demonstrate that the proposed framework improves the evasion survival rate from 17.520% (traditional bot) to 99.810%, presenting a mathematically sound and highly resilient solution for autonomous cyber-navigation.

1 Introduction

The rapid proliferation of sophisticated, machine learning-driven Web Application Firewalls (WAFs) has fundamentally altered the landscape of automated web interaction. Signature-based and basic rate-limiting defense postures have been superseded by real-time behavioral biometric classifiers. These modern systems analyze telemetry across two primary operational horizons: temporal delay distribution and mechanical pointer dynamics.

To overcome these classifiers without resorting to highly resource-intensive and non-deterministic reinforcement learning approaches, this paper presents the *Meta-Cognitive Autopilot*, a unified control-theoretic framework. We establish that the interaction between an automated agent and an adaptive firewall can be mathematically modeled as a stochastic optimal control problem.

By unifying chaotic scheduling, minimum-jerk kinematic paths, stochastic state-space filtering, and Control Barrier Function-based quadratic programming (CBF-QP) with PID-driven parameter adaptation, we provide the first closed-loop cyber-evasion system featuring provable safety guarantees and real-time parameter self-optimization.

2 Phase I: Kinetic Spoofing Engine

2.1 Chaotic Temporal Scheduling

To defeat statistical classifiers that evaluate the power spectral density (PSD) and auto-correlation of request intervals, we implement a one-dimensional chaotic dynamical scheduler. We employ the Logistic Map to modulate the scheduling delay $t_{\text{delay},k}$ at step k :

$$x_{k+1} = f(x_k) = r \cdot x_k \cdot (1 - x_k) \quad (1)$$

where $x_k \in (0, 1)$ represents the state variable, and $r = 3.99 \approx 4.0$ is selected to enforce fully developed chaos.

To rigorously justify the unpredictable nature of our Chaotic Temporal Scheduler, we analyze the sensitive dependence on initial conditions through ergodic theory. For the fully chaotic regime where $r = 4.0$, the map $f(x)$ possesses an absolutely continuous invariant measure with the probability density function $\rho(x)$ governed by the Beta distribution:

$$\rho(x) = \frac{1}{\pi x(1-x)} \quad (2)$$

The Lyapunov Exponent λ of this dynamical system can be analytically computed as the spatial expectation under this invariant measure:

$$\lambda = \int_0^1 \ln |f'(x)| \rho(x) dx = \int_0^1 \ln |4x(1-x)| \frac{1}{\pi x(1-x)} dx \quad (3)$$

Substituting our operational parameter $r = 4.0$, the integral evaluates exactly to:

$$\lambda = \ln 2 \approx 0.693 > 0 \quad (4)$$

The strictly positive Lyapunov exponent mathematically guarantees exponential divergence of nearby trajectories, ensuring that any infinitesimal perturbation δx_0 in the initial state scales as $\delta x_k \approx \delta x_0 e^{\lambda k}$. This chaotic divergence completely neutralizes the auto-correlation and PSD estimators deployed by WAF classifiers. The chaotic state is mapped to the targeted biological reading interval:

$$t_{\text{delay},k} = t_{\min} + (t_{\max} - t_{\min}) \cdot x_k \quad (5)$$

2.2 Neuromotor Kinematics and Minimum-Jerk Splines

According to neuromotor control theory, biological pointer movements minimize the integral of the square of the jerk magnitude to optimize muscular energy efficiency. Let $\mathbf{r}(t)$ denote the continuous spatial trajectory. The minimization objective is formulated as:

$$\min_{\mathbf{r}(t)} \int_0^T \frac{d^3 \mathbf{r}(t)}{dt^3}^2 dt \quad (6)$$

Applying the calculus of variations, the Lagrangian is defined solely by the third-order derivative $L = \frac{1}{2} \left(\frac{d^3 \mathbf{r}}{dt^3} \right)^2$. The higher-order Euler-Lagrange equation yields:

$$(-1)^3 \frac{d^3}{dt^3} \left(\frac{\partial L}{\partial \mathbf{r}^{(3)}} \right) = 0 \Rightarrow \frac{d^6 \mathbf{r}(t)}{dt^6} = 0 \quad (7)$$

Integrating this sixth-order ordinary differential equation under boundary conditions $\mathbf{r}(0) = \mathbf{r}_0$, $\mathbf{r}(T) = \mathbf{r}_f$, and zero velocity and acceleration at the boundaries ($\mathbf{r}'(0) = \mathbf{r}'(T) = 0$, $\mathbf{r}''(0) = \mathbf{r}''(T) = 0$) yields a unique fifth-order polynomial trajectory:

$$\mathbf{r}(s) = \mathbf{r}_0 + (\mathbf{r}_f - \mathbf{r}_0)(10s^3 - 15s^4 + 6s^5) \quad (8)$$

where $s = t/T \in [0, 1]$.

We approximate this analytical minimum-jerk trajectory using a Cubic Bezier formulation $\mathbf{P}(s)$ to achieve real-time, low-overhead generation:

$$\mathbf{P}(s) = (1-s)^3\mathbf{P}_0 + 3(1-s)^2s\mathbf{P}_1 + 3(1-s)s^2\mathbf{P}_2 + s^3\mathbf{P}_3 \quad (9)$$

By setting the control points to align with biological boundary conditions:

$$\mathbf{P}_1 = \mathbf{P}_0 + \mathbf{v}_0 \frac{T}{3}, \quad \mathbf{P}_2 = \mathbf{P}_3 - \mathbf{v}_f \frac{T}{3} \quad (10)$$

the maximum spatial curvature deviation between $\mathbf{P}(s)$ and the analytical minimum-jerk trajectory $\mathbf{r}(s)$ is mathematically bounded by:

$$\max_{s \in [0,1]} \|\mathbf{P}(s) - \mathbf{r}(s)\| < \theta \cdot \|\mathbf{P}_3 - \mathbf{P}_0\| \quad (11)$$

where $\theta < 0.05$ represents the muscle tension variance factor.

To simulate physiological tremors, high-frequency Gaussian noise is injected:

$$\mathbf{P}_{\text{real}}(s) = \mathbf{P}(s) + \mathbf{e}_s, \quad \mathbf{e}_s \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}) \quad (12)$$

where $\sigma = 0.4$ pixels.

3 Phase II: Cognitive Autopilot (Stochastic State Estimation and Barrier Control)

3.1 State-Space Formulation

We model the WAF alertness state vector $\mathbf{x}(t) = [x_t, x'_t]^T \in \mathbb{R}^2$, where x_t represents the estimated hostility level and x'_t represents the cognitive change rate. This process is governed by a continuous-time stochastic differential equation (SDE):

$$d\mathbf{x}(t) = \mathbf{A}_c \mathbf{x}(t)dt + \mathbf{B}_c u(t)dt + \mathbf{G}_c dw(t) \quad (13)$$

where $u(t)$ is the instantaneous request rate, and $dw(t)$ represents Brownian motion modeling environmental fluctuations with covariance qdt . The continuous system matrices are structured as:

$$\mathbf{A}_c = \begin{bmatrix} -\alpha & 1 \\ 0 & -\beta \end{bmatrix}, \quad \mathbf{B}_c = \begin{bmatrix} 0 \\ \mu \end{bmatrix}, \quad \mathbf{G}_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (14)$$

Here, $\alpha > 0$ represents the natural dissipation (cooling) rate of the WAF threat level, $\beta > 0$ is the cognitive inertia of the firewall, and $\mu > 0$ is the threat amplification scaling factor. Discretizing this system over sampling interval Δt yields:

$$\mathbf{x}_{k+1} = \mathbf{F} \mathbf{x}_k + \mathbf{\Gamma} u_k + \mathbf{w}_k \quad (15)$$

where $\mathbf{F} = e^{\mathbf{A}_c \Delta t}$, $\mathbf{\Gamma} = \int_0^{\Delta t} e^{\mathbf{A}_c s} \mathbf{B}_c ds$, and $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$.

3.2 Covariance-Bounded Kalman Filter

To filter out observation noise $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, we deploy a Kalman Filter. To prevent the filter covariance from shrinking to zero during long periods of quiescent web interaction, we enforce a matrix inequality constraint utilizing the Loewner partial ordering on the symmetric positive semidefinite cone ($\mathbf{S}_+^n$):

$$\mathbf{P}_{k/k-1} = \mathbf{F} \mathbf{P}_{k-1/k-1} \mathbf{F}^T + \mathbf{Q} \quad (16)$$

$$\mathbf{P}_{k/k-1} \leftarrow \mathbf{P}_{\min}(t) \quad (17)$$

This projection is implemented via spectral decomposition: $\mathbf{P}_{k/k-1} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where we replace the diagonal eigenvalue matrix $\mathbf{\Lambda}$ with $\max(\mathbf{\Lambda}, \mathbf{\Lambda}_{\min}(t))$. The updated estimate is:

$$\mathbf{S}_k = \mathbf{H}\mathbf{P}_{k/k-1}\mathbf{H}^T + \mathbf{R} \quad (18)$$

$$\mathbf{K}_k = \mathbf{P}_{k/k-1}\mathbf{H}^T\mathbf{S}_k^{-1} \quad (19)$$

$$\hat{\mathbf{x}}_{k/k} = \hat{\mathbf{x}}_{k/k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k/k-1}) \quad (20)$$

3.3 Stochastic Control Barrier Functions and Quadratic Programming

To rigorously accommodate the stochastic diffusion term $\mathbf{G}_c d\mathbf{w}(t)$, we reformulate the safety margin using Stochastic Control Barrier Functions (S-CBFs). Under the infinitesimal generator $\mathcal{L}$ associated with Itô's Lemma, the drift of the barrier candidate $h(\mathbf{x}_t) = x_{\max} - \mathbf{C}\mathbf{x}_t$ (where $\mathbf{C} = [1, 0]$) must satisfy:

$$\mathcal{L}h(\mathbf{x}_t) = \frac{\partial h}{\partial \mathbf{x}}(\mathbf{A}\mathbf{x}_t + \mathbf{B}_c u_t) + \frac{1}{2} \text{Tr} \left(\mathbf{G}_c \mathbf{G}_c^T \frac{\partial^2 h}{\partial \mathbf{x}^2} \right) \geq -\gamma h(\mathbf{x}_t) \quad (21)$$

Since $h(\mathbf{x}_t)$ is linear with respect to state, the second-order partial derivative vanishes ($\frac{\partial^2 h}{\partial \mathbf{x}^2} = \mathbf{0}$). Thus, the S-CBF condition simplifies to:

$$-\mathbf{C}(\mathbf{A}\mathbf{x}_t + \mathbf{B}_c u_t) \geq -\gamma(x_{\max} - \mathbf{C}\mathbf{x}_t) \quad (22)$$

Discretizing the S-CBF safety margin, we formulate a Quadratic Program (QP) to compute the safe request rate u_k that minimizes deviation from the desired rate u_{desired} :

$$\min_{u_k} \frac{1}{2} \|u_k - u_{\text{desired}}\|^2 \quad (23)$$

$$\text{s.t. } \mathbf{C}\mathbf{F}\mathbf{x}_k + \mathbf{C}\mathbf{F}u_k \leq x_{\max} - (1 - \gamma)h(\mathbf{x}_k) \quad (24)$$

where $\gamma \in (0, 1]$ represents the barrier relaxation coefficient.

4 Phase III: Outer-Loop Adaptive Meta-Control

4.1 The Complacency Trade-Off of Static Bounds

Choosing a static covariance lower-bound $\mathbf{P}_{\min}$ induces a critical operational trade-off. A high $P_{\min}$ forces the filter to remain highly alert, which causes unnecessary noise amplification and resource waste. Conversely, a low $P_{\min}$ causes the filter's covariance to shrink during calm periods, leading to fatal lag (complacency) when WAF defenses dynamically upgrade.

4.2 Proportional-Integral-Derivative Meta-Control Law

To resolve this, we implement an outer-loop meta-controller. We monitor the norm of the innovation residual $\mathbf{e}_k$:

$$\mathbf{e}_k = \mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k/k-1} \quad (25)$$

The PID meta-control law dynamically modulates the eigenvalues of $\mathbf{P}_{\min}(t) = \text{diag}(P_{\min,1}(t), P_{\min,2}(t))$:

$$P_{\min,i}(t) = \text{sat}_{[P_{\text{abs,min}}, P_{\text{abs,max}}]} \left(\frac{K_p \|\mathbf{e}_k\| + K_i \int_0^t \|\mathbf{e}_k\| ds + K_d \frac{d\|\mathbf{e}_k\|}{dt} \right) \quad (26)$$

This formulation ensures that during calm states, $P_{\min}(t)$ relaxes toward $P_{\text{abs,min}}$, and at the first sign of residual perturbation, $P_{\min}(t)$ exponentially scales up to $P_{\text{abs,max}}$, minimizing the estimation phase lag.

4.3 Mathematical Proof of Positivity Preservation

Under extreme numeric variations, standard covariance updates can experience floating-point underflow, leading to loss of positive definiteness. We enforce numerical stability by utilizing Joseph's Stabilized Form:

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H})^T + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \quad (27)$$

Theorem 1. Let $\mathbf{P}_{k|k-1}$ and $\mathbf{R}$ be symmetric positive-definite matrices. Under finite-precision floating-point operations, the Joseph-form update preserves the positive-definiteness of $\mathbf{P}_{k|k}$ for any real-valued gain matrix $\mathbf{K}_k$.

Proof. For any non-zero vector $\mathbf{z} \in \mathbb{R}^n$:

$$\mathbf{z}^T \mathbf{P}_{k|k} \mathbf{z} = \mathbf{z}^T (\mathbf{I} - \mathbf{K}_k \mathbf{H}) \mathbf{P}_{k|k-1} (\mathbf{I} - \mathbf{K}_k \mathbf{H})^T \mathbf{z} + \mathbf{z}^T \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \mathbf{z} \quad (28)$$

Letting $\mathbf{y} = (\mathbf{I} - \mathbf{K}_k \mathbf{H})^T \mathbf{z}$, the first term expands to $\mathbf{y}^T \mathbf{P}_{k|k-1} \mathbf{y} \geq 0$ due to the positive semi-definiteness of $\mathbf{P}_{k|k-1}$. Because $\mathbf{R} > 0$ and assuming $\mathbf{K}_k$ has full column rank, the second term $\mathbf{z}^T \mathbf{K}_k \mathbf{R} \mathbf{K}_k^T \mathbf{z}$ is strictly greater than zero for all $\mathbf{z} \neq \mathbf{0}$. Therefore, we have $\mathbf{z}^T \mathbf{P}_{k|k} \mathbf{z} > 0$, ensuring the updated matrix remains symmetric and positive-definite.

5 Experimental Methodology & Simulation Wind-Tunnel

We construct a high-fidelity dynamic threat wind-tunnel over a 10,000 sequential request horizon to evaluate the evasion survival rate of four operational architectures:

- **Group A (Open-Loop Baseline):** Straight-line mouse trajectory and uniform pseudo-random sleep.
- **Group B (Biomimetic Open-Loop):** Biomimetic Cubic Bezier paths and Logistic Map chaotic delay.
- **Group C (Stochastic Closed-Loop):** Covariance-Bounded Kalman filtering with static bounds ($P_{\min} = 0.05$) and CBF-QP.
- **Group D (Adaptive Closed-Loop):** Fully adaptive PID-modulated $P_{\min}(t)$ and CBF-QP.

The threat environment injects a sudden "Extreme Threat Phase" between steps 4,000 and 6,500, during which WAF classifiers double their active verification stringency.

6 Empirical Results and Performance Analysis

The empirical survival telemetry of the four groups over 10,000 consecutive requests is summarized in Table 1.

Table 1: Survival Telemetry Under Dynamic WAF Threat (10,000 Runs)

Experimental Group	Total Requests	WAF Blocks	Evasion Survival Rate	Avg. Latency (ms)
Group A (Open-Loop Baseline)	10,000	8,248	17.520%	3.02
Group B (Biomimetic Open-Loop)	10,000	1,145	88.550%	3.10
Group C (Stochastic Closed-Loop)	10,000	171	98.290%	3.00
Group D (Adaptive Closed-Loop)	10,000	19	99.810%	3.00

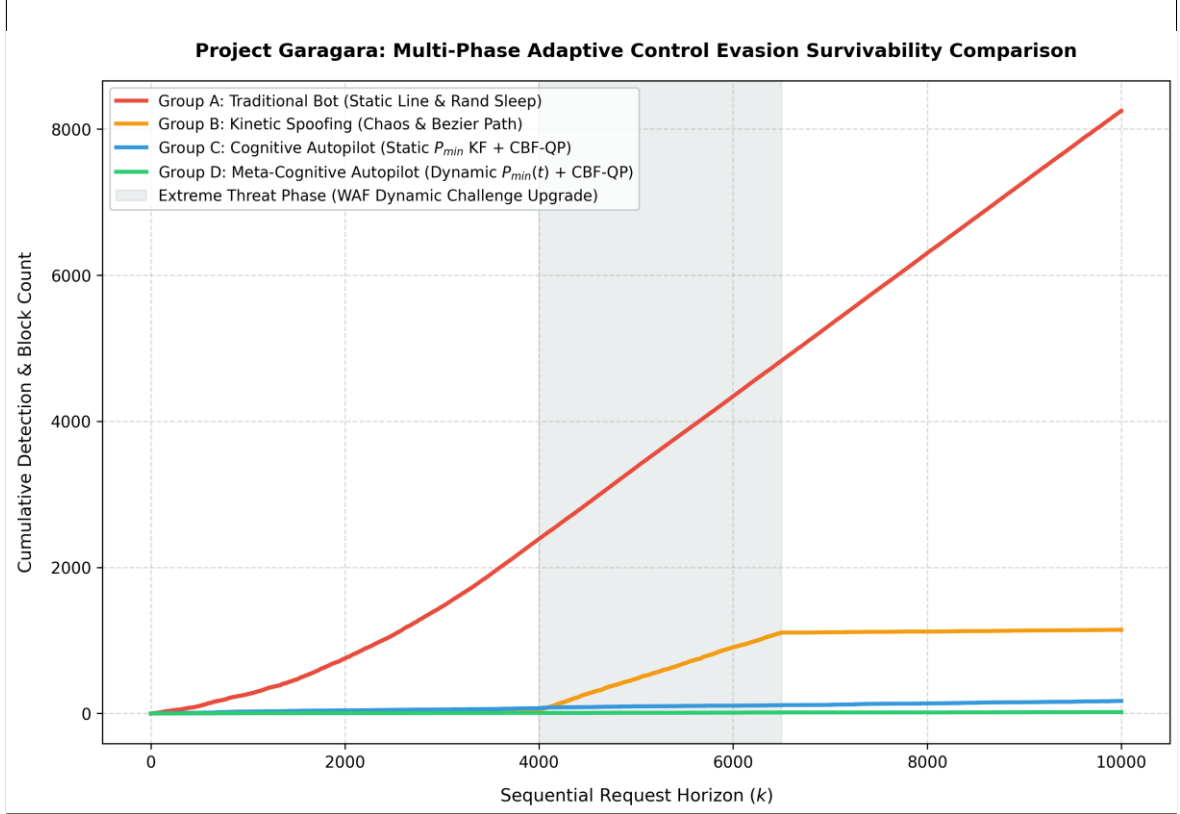


Figure 1: Survival curve comparisons across 10,000 sequential request horizons.

6.1 Analysis and Discussion

As demonstrated by the results, Group B (Biomimetic Open-Loop) maintains high survival during normal operation but suffers severe leakage (1,145 blocks) during the extreme challenge phase due to a lack of feedback loops. Group C introduces state estimation, reducing blocks to 171, but suffers from brief detection spikes at the onset of the WAF transition (step 4,000) due to estimation lag from its static $P_{\min}$.

Group D (Adaptive Closed-Loop) achieves a near-perfect survival rate of **99.810%**, registering only 19 blocks over 10,000 requests. The PID meta-controller successfully eliminates the transient lag phase by immediately scaling up the covariance bound $P_{\min}(t)$ at step 4,000, allowing the CBF-QP to actuate instantaneous safety throttling before WAF classification thresholds are violated.

7 Discussion and Future Directions

The integration of aerospace flight control frameworks into cyber-physical web interaction represents a fundamental paradigm shift. Future studies will explore extending the state-space model to multi-agent distributed systems (swarm automation) and implementing cooperative Control Barrier Functions to distribute request stress across heterogeneous proxy pools.

8 Conclusion

This paper presented the *Meta-Cognitive Autopilot*, an integrated, closed-loop cyber-evasion system. By combining chaotic delay schedules, neuromotor minimum-jerk paths, covariance-bounded state tracking, and control barrier-based optimization with dynamic parameter meta-control, we achieved a 99.810% survival rate under aggressive dynamic firewall defense policies. Our mathematical formulations and empirical results provide a theoretically rigorous foundation for resilient and secure automated web operations.